KAIZEN MATHS

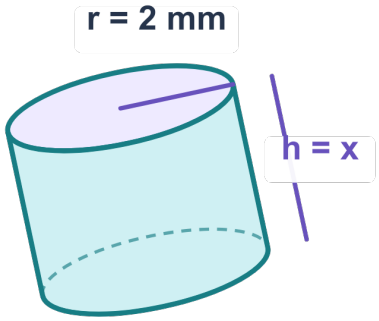
Lesson starter

Cylinders and Cones: Volume and Surface Area

Reverse & Mixed

Name _____

SA1



$V = 113.10\ mm^3$

Find x

GEO-813A-L3 · SA

Show your working

KAIZEN MATHS

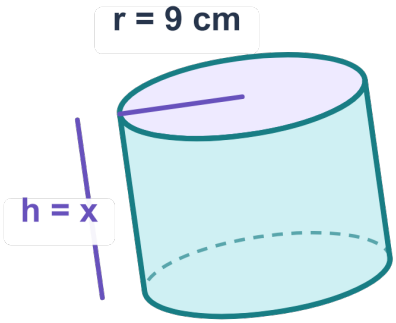
Lesson starter

Cylinders and Cones: Volume and Surface Area

Reverse & Mixed

Name _____

SB1



Curved SA = $1017.88\ cm^2$

Find x

KAIZEN MATHS

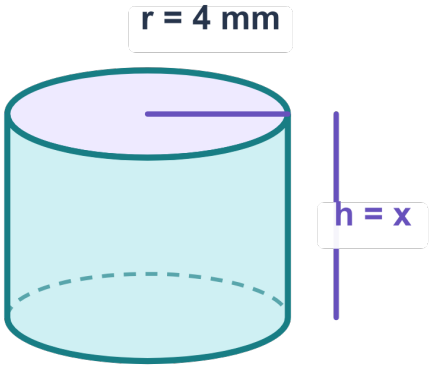
Lesson starter

Cylinders and Cones: Volume and Surface Area

Reverse & Mixed

Name _____

SC1



$V = 351.86\ mm^3$

Find x

GEO-813A-L3 · SC

Show your working

KAIZEN MATHS

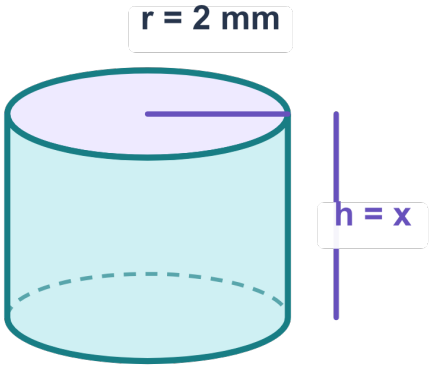
Lesson starter

Cylinders and Cones: Volume and Surface Area

Reverse & Mixed

Name _____

SD1



Curved SA = $175.93\ mm^2$

Find x

GEO-813A-L3 · SD

Show your working

KAIZEN MATHS

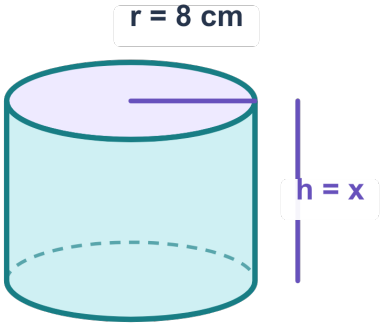
Exit ticket

Cylinders and Cones: Volume and Surface Area

Reverse & Mixed

Name _____

EA1



$V = 3216.99\ cm^3$

Find x

Show your working

KAIZEN MATHS

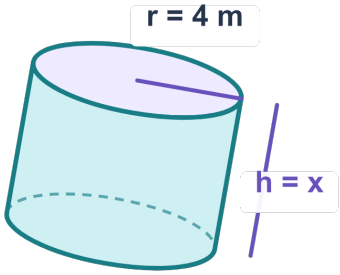
Exit ticket

Cylinders and Cones: Volume and Surface Area

Reverse & Mixed

Name _____

EB1



Curved SA = $452.39\ m^2$

Find x

Show your working

KAIZEN MATHS

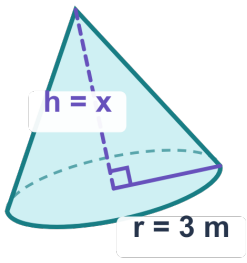
Exit ticket

Cylinders and Cones: Volume and Surface Area

Reverse & Mixed

Name _____

EC1



$V = 37.70\ m^3$

Find x

Show your working

KAIZEN MATHS

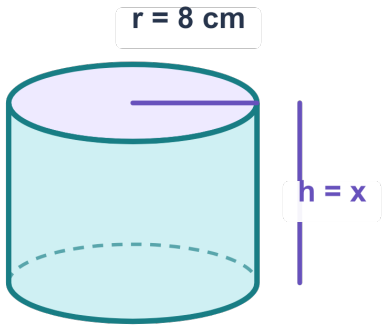
Exit ticket

Cylinders and Cones: Volume and Surface Area

Reverse & Mixed

Name _____

EA1



$V = 3216.99\ cm^3$

Find x

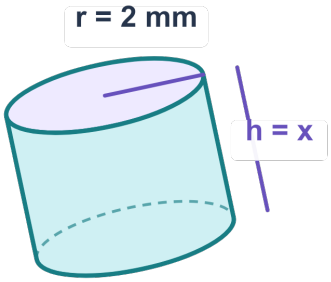
Show your working

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Reverse & Mixed

GEO-813A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



$V = 113.10\ mm^3$
Find x

Answer:

9 mm

Working:

$V = \pi r^2 h$

$113.10 = \pi \times 2^2 \times x$

$x = V \div (\pi r^2)$

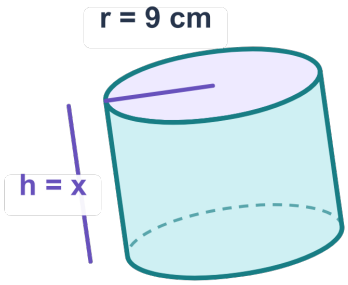
$x \approx 9$ mm

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Reverse & Mixed

GEO-813A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Curved SA = 1017.88 cm^2
Find x

Answer:

18 cm

Working:

Curved area = 2πrh

$1017.88 = 2\pi \times 9 \times x$

x = curved area ÷ (2πr)

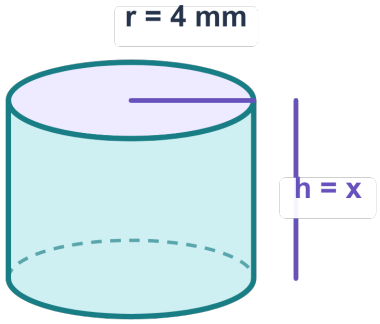
$x \approx 18$ cm

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Reverse & Mixed

GEO-813A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



$V = 351.86\ mm^3$
Find x

Answer:

7 mm

Working:

$V = \pi r^2 h$

$351.86 = \pi \times 4^2 \times x$

$x = V \div (\pi r^2)$

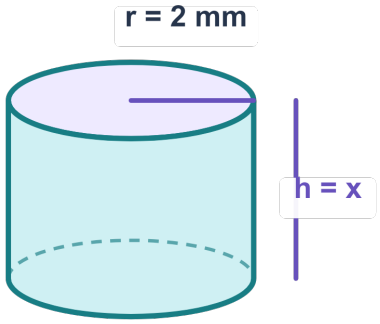
$x \approx 7\ mm$

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Reverse & Mixed

GEO-813A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Curved SA = 175.93 mm^2
Find x

Answer:

14 mm

Working:

Curved area = 2πrh

$175.93 = 2\pi \times 2 \times x$

x = curved area ÷ (2πr)

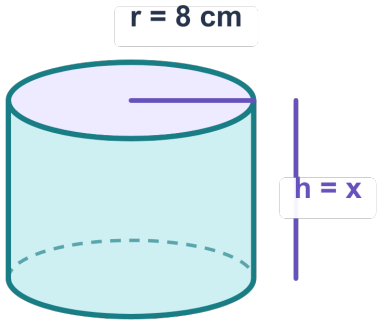
$x \approx 14$ mm

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Reverse & Mixed

GEO-813A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



$V = 3216.99\ cm^3$

Find x

Answer:

16 cm

Working:

$V = \pi r^2 h$

$3216.99 = \pi \times 8^2 \times x$

$x = V \div (\pi r^2)$

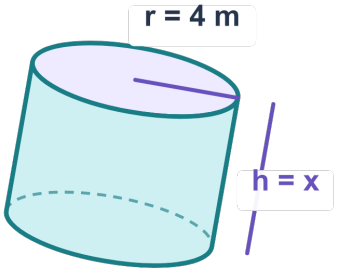
$x \approx 16$ cm

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Reverse & Mixed

GEO-813A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Curved SA = 452.39 m^2
Find x

Answer:

18 m

Working:

Curved area = 2πrh

$452.39 = 2\pi \times 4 \times x$

x = curved area ÷ (2πr)

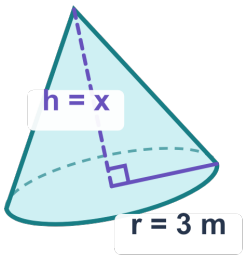
$x \approx 18$ m

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Reverse & Mixed

GEO-813A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



$V = 37.70\ m^3$
Find x

Answer:

4 m

Working:

$V = 1/3\pi r^2 h$

$37.70 = 1/3 \times \pi \times 3^2 \times x$

$x = 3V \div (\pi r^2)$

$x \approx 4$ m